



Institutional-Grade Algorithmic Trading Platform

Version 1.0 (V1) & 2.0 (V2) - BRD

Business Requirements Document (BRD)
Comprehensive Architecture, Trading Logic & Feature Analysis.

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1. Executive Summary

Project Goal: To develop a fully autonomous, high-frequency intraday trading bot for the Indian Financial Markets (NSE/BSE), combining quantitative technical indicators with cutting-edge Machine Learning and real-time Macro Sentiment Analysis.

QuantAI is designed to execute trades systematically without human emotional interference. By leveraging the Fyers API for high-speed order execution and market data polling, the system analyzes price action using multi-timeframe indicators and predictive AI models to generate high-probability BUY/SELL signals for index options (NIFTY & BANKNIFTY).

1.1 Core Objectives

- Automated Execution:** Zero-touch order placement with automatic stop-loss, trailing stop-loss, and take-profit mechanisms.
- AI-Driven Edge:** Utilize predictive Machine Learning models (Random Forest Classifier) to score market probability based on historical patterns.
- News Sentiment Integration (V2):** Analyze real-time financial news using Natural Language Processing (VADER NLP) to gauge overall market bias and translate it to Telugu for localized access.
- Comprehensive Backtesting:** Provide a robust engine to simulate strategies against historical tick and candle data before live deployment.

2. Architecture & Technology Stack

The system is built on a modern, decoupled architecture separating the fast-execution Python backend from the interactive monitoring frontend.

Frontend Dashboard (User Interface)

Next.js 14

React

TailwindCSS

TypeScript

Recharts

Framer Motion

- A sleek, dark-themed "glassmorphism" dashboard for institutional-grade visual appeal.
- Real-time WebSocket/Polling connectivity for live PnL, active trades, and AI confidence meters.
- Bilingual (English & Telugu) auto-rotating news ticker for real-time macroeconomic sentiment tracking.

Backend & Trading Engine

- **API Bridge:** Uvicorn-powered FastAPI server acting as the central hub.
- **Trading Loop:** Asynchronous thread managing continuous market data polling and execution logic.
- **Sentiment Analyzer (V2):** Background daemon fetching RSS feeds, utilizing vaderSentiment for scoring, and deep-translator for localization.

3. Key Features & Modules

3.1 Technical Indicators & Strategy Logic

The bot computes sophisticated mathematical indicators in real-time. The confluence of these indicators creates the underlying technical edge.

Indicator	Purpose in QuantAI
CPR (Central Pivot Range)	Identifies strong support/resistance zones and daily market trend direction. Price breaking CPR levels acts as a primary trigger.
Squeeze Momentum	Detects periods of low volatility (consolidation) to predict explosive breakouts (Bollinger Bands inside Keltner Channels).
EMA Crossovers	Uses 9 EMA and 21 EMA crossovers to confirm short-term momentum shifts and ride trends.
RSI (Relative Strength Index)	Measures overbought/oversold conditions to avoid entering trades at the peak of a move.

3.2 AI & Machine Learning Module (Predictive Edge)

- **Model Architecture:** Random Forest Classifier (Ensemble Learning).
- **Feature Engineering:** Trained on historical tick/candle data, deriving 15+ features from technical indicators (Price distance from EMA, CPR state, Squeeze fired, etc.).
- **Output:** Generates a probability array indicating the likelihood of a bullish (Call) or bearish (Put) move in the next 15-30 minutes.
- **Retraining Pipeline:** Automated daily retraining scripts (`daily_ai_retrain.py`) to adapt the model weights to recent market regimes.

3.3 Live News Sentiment (V2 Feature)

A critical addition implemented in V2 to prevent the technical bot from trading against sudden macroeconomic trends or global news shocks.

- **Data Sources:** High-frequency RSS polling from Economic Times, Moneycontrol, and LiveMint.

- **NLP Processing:** VADER NLP analyzes headlines to compute a compound sentiment score ranging from -1 (Crash) to +1 (Euphoria).
- **Localization:** Headlines are dynamically translated to Telugu for user accessibility using Google Translate APIs.
- **Non-Blocking Design:** Fetches data in an asynchronous daemon thread to ensure 0ms latency impact on the core trading engine.

4. Backtesting Engine

A standalone, offline engine critical for strategy validation located in `backtesting_engine/run.py`.

- **Historical Simulation:** Replays past market data bar-by-bar, accurately simulating fill prices, slippage, and brokerage costs.
- **Metrics Calculated:**
 - **Total Return (%):** Absolute growth of initial capital.
 - **Max Drawdown:** The largest peak-to-trough drop in equity (risk measure).
 - **Sharpe Ratio:** Risk-adjusted return metric.
 - **Win/Loss Ratio & Profit Factor:** Consistency metrics.
- **Validation:** Ensures that the AI model and indicator logic have a statistical edge before risking real capital in live markets.

5. Risk Management Protocol

The system is strictly bound by institutional risk management rules to prevent blow-ups.

- **Hard Stop Loss:** Every trade is placed with an immutable hard stop-loss at the broker level.
- **Trailing Stop Loss (TSL):** Once a trade is profitable, the stop-loss is dynamically trailed to lock in gains.
- **Daily Max Loss:** The bot halts trading for the day if cumulative losses exceed the user-defined threshold.
- **Overtrading Prevention:** Limits the maximum number of trades per symbol per day.

6. Future Enhancements (V3 Roadmap)

- **Options Decoding (Greeks):** Real-time calculation of Delta, Theta, Vega, and Gamma for advanced hedging strategies (e.g., Iron Condors, Straddles).
- **Multi-Broker Support:** Architecting an adapter pattern to extend execution capabilities to Zerodha (Kite Connect), Upstox, and Angel One.
- **Deep Learning Models:** Upgrading the predictive engine from Random Forest to LSTM (Long Short-Term Memory) neural networks for better sequence prediction in time-series data.
- **Telegram/WhatsApp Integration:** Instant push notifications for trade executions, PnL updates, and critical risk warnings directly to the user's mobile device.

